ESP32 CSI Patient Detection — Setup & Data/Training Plan

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Overview

The deployed system uses ESP32 Wi-Fi channel state information (CSI) to detect patient presence, bed occupancy, posture, bed exits and falls, with no camera in the room. Cameras are used only during a bounded collection phase, where pose estimation turns video into time-stamped labels that train the CSI models.

CSI is the per-subcarrier amplitude and phase of each received Wi-Fi packet. A body in the room reflects and absorbs the signal, so movement, posture and even chest motion change the CSI pattern across subcarriers and over time.

Output	Classes or value	Clinical use	Difficulty
Room occupancy	empty / 1 person / 2+ people	Rounding verification, staff presence	Easy
Bed occupancy	in bed / out of bed	Bed-exit alerts for fall-risk patients	Easy to moderate
Posture and activity	lying, sitting on bed edge, standing, walking	Pre-exit warning: sitting up usually precedes an exit	Moderate
Fall event	fall / no fall	Faster response to unwitnessed falls	Hard (rare events)
Respiration	breaths per minute while lying still	Early deterioration warning	Hard, research-grade

The main technical risk is that CSI models are highly environment-specific: a model trained in one room often degrades in another, or after furniture moves. The plan below is built around that risk, with collection across many rooms and leave-one-room-out evaluation.

Hardware setup

Each room gets one dedicated ESP32 transmitter and three ESP32 receivers with external antennas, all wired by USB to a small logging PC. This avoids depending on hospital Wi-Fi, which you cannot control, and gives three radio paths through the zones that matter.

Component	Recommended part	Qty per room	Notes
Transmitter (TX)	ESP32-S3 board with u.FL antenna connector (e.g. ESP32-S3-DevKitC-1U)	1	Sends fixed-rate packets on one fixed channel
Receivers (RX)	Same board	3	Each logs CSI from the TX's packets
Antennas	2.4 GHz dipole + u.FL-to-SMA pigtail	4	External antennas outperform the directional PCB antenna
Logging host	Mini PC or Raspberry Pi 5 + powered USB hub	1	Timestamps CSI and camera frames on one clock
Mounts	Clamp or adhesive mounts, cable raceway	4	Fixed, labelled positions; photograph every install

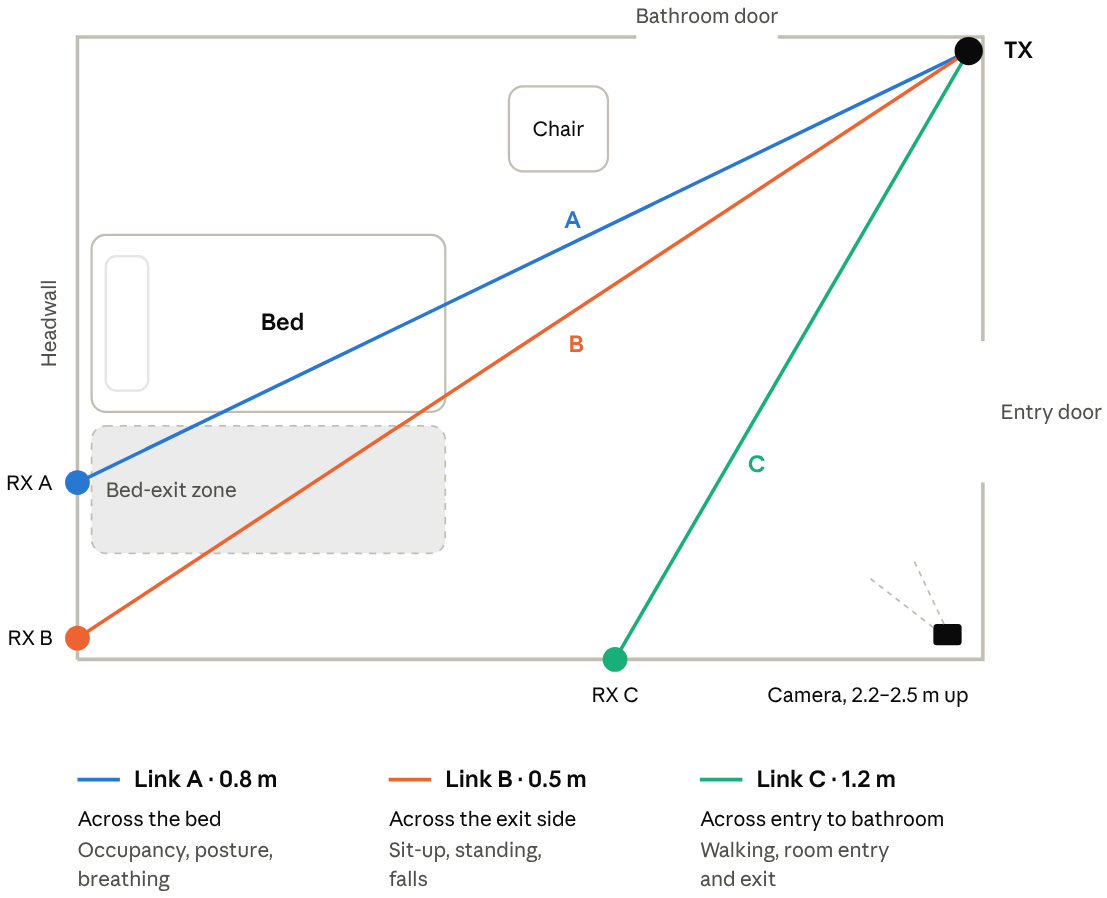
Why a dedicated transmitter. Espressif's esp-csi docs rate a dedicated packet sender as the most accurate way to get CSI. Hospital access points change channels, roam clients and carry unpredictable traffic, all of which look like motion to a CSI model.

Where the three links go. Each TX→RX line should cut through one zone of interest:

- **Link A, across the bed at mattress height (~0.8 m):** occupancy, posture, breathing.
- **Link B, across the bed-exit side, mounted low (~0.5 m):** sitting up, standing, falls near the floor.
- **Link C, across the door and bathroom path (~1.2 m):** walking, room entry and exit.

Each link cuts through one zone: the bed, the bed-exit side, the door path

Top-down view of one patient room, not to scale



room layout · 1 transmitter, 3 receivers, 1 camera

The transmitter sits high in the corner opposite the headwall, and the camera in the other far corner; all four ESP32s and the camera connect by USB to one logging PC in the room.

Radio settings. One fixed 2.4 GHz channel (1, 6 or 11, chosen with hospital IT), HT20 to start, fixed TX power, 100 packets per second. Log channel, bandwidth, RSSI, noise floor and the ESP's microsecond timestamp with every packet.

Getting data off the boards. Stream over USB serial at 921600 baud or higher, not over Wi-Fi. Uploading over Wi-Fi adds traffic on the very channel you are sensing.

Chip choice. ESP32-S3 is the default: most examples target it, including Espressif's console_test collection tool. Espressif lists CSI support across the ESP32, S2, C3, S3, C5, C6 and C61. The ESP32-C5 is worth a second-phase pilot for its 5 GHz band and the esp-crab phase-sync solution Espressif ships for it; 5 GHz resolves small motion like breathing better but has shorter range.

Camera ground truth

One ceiling-corner camera per room records to the same logging PC as the ESP32s, and pose estimation runs locally to turn video into labels. Only labels and skeletons leave the room; raw video is kept encrypted for a short audit window, then deleted.

Option	Strengths	Weaknesses	Use it for
RGB camera with IR night mode	Cheapest; pose models are most accurate on RGB	Most identifiable; needs IR light at night	Pilot rooms, staff-only rehearsal sessions
Depth camera (e.g. Orbbec Femto Bolt, Intel RealSense D455)	Works in the dark; far less identifiable; gives height above floor	Costs more; fewer pretrained models	Patient rooms, falls, in-bed posture

Placement. Mount 2.2 to 2.5 m high in the corner facing the headwall, so one view covers the bed, the bed-exit side and the door. Never place a camera inside the bathroom; bathroom events are labelled by staff notes or left unlabelled.

Synchronization. Both streams are timestamped by the logging PC's clock, so they share one time base. Verify it every session: an ESP32 blinks an LED in a coded pattern visible to the camera while logging each blink in the CSI stream. The offset between the two should stay under 50 ms, about 1.5 video frames at 30 fps.

Turning video into labels.

1. Detect and track people and 17 body keypoints per frame with a pose model: YOLO26-pose or RTMPose from MMPose.
2. Draw the bed outline once per install, and again whenever the bed moves.
3. Derive labels by rule: person count; in bed when the hip keypoints fall inside the bed outline; posture from torso angle and hip height; fall candidates from a fast drop in hip height that ends near the floor.
4. Send every fall candidate, every bed-exit and a random 5% of windows to human review in CVAT or Label Studio.

Known gaps. Pose models struggle under blankets, so in-bed posture labels are weaker than in-bed versus out-of-bed. Cameras cannot give reliable breathing rate; if respiration is in scope, add a chest respiratory belt or, with approval, the bedside monitor's respiratory-rate feed.

Licensing. YOLO26 is AGPL-3.0, which requires open-sourcing derivative work unless you buy an enterprise license. MMPose is Apache-2.0, so it is the safer default if this may become a product.

Software and data pipeline

Flash Espressif's esp-csi `csi_send` example on the transmitter and `csi_recv` on the receivers, on a current ESP-IDF 5.x release. A Python logger on the host reads each serial port, the camera process writes frames, and both stamp data with the same clock (chrony-synced, nanosecond monotonic time).

Watch the serial bandwidth. Text CSV at 100 packets per second nearly saturates a 921600-baud link. Switch the receiver output to binary or base64, or use the S3's native USB port.

Storage layout, one folder per session:

- `csi/rx{1,2,3}.parquet`: host timestamp, ESP timestamp, packet sequence number, RSSI, noise floor, channel, bandwidth, raw CSI as int8 I/Q pairs.
- `video/`: encrypted camera files, deleted after the audit window.
- `pose.parquet`: per-frame keypoints, scores and track IDs.
- `labels.parquet`: the label table below at 10 Hz.
- `session.json`: room, install ID, bed position, door state, people present, consent ID, sync offset, notes.

Raw CSI runs to roughly 8 GB per room per day at 3 receivers and 100 Hz (approximate, before compression).

Label field	Values	Rate	Source
occupancy_count	0, 1, 2+	10 Hz	Pose tracker
patient_in_bed	yes, no, unknown	10 Hz	Hip keypoints inside bed outline
posture	lying, sitting on bed edge, sitting in chair, standing, walking, on floor, unknown	10 Hz	Rules + human review
others_present	yes, no	10 Hz	Tracks other than the patient's

Label field	Values	Rate	Source
event	bed_exit, fall, room_entry, room_exit	Timestamped	Rules + mandatory review
respiration_bpm	breaths per minute	1 Hz	Chest belt or monitor (optional)
label_quality	auto, reviewed, disputed	Per window	Review tool

The patient's track is picked as the person in bed at session start, confirmed by a reviewer. `others_present` matters because staff and visitors move the CSI as much as the patient does, and the model must learn to tell them apart.

Approvals, privacy and safety

No camera goes into a patient room until the IRB has approved the protocol and each patient, or their legal representative, has consented. This section assumes a US hospital; video of patients is protected health information under HIPAA, since full-face images are one of its 18 identifiers.

- [ ] IRB protocol covering cameras, CSI, retention and withdrawal; expect written consent, not a waiver
- [ ] Consent or notice for staff, plus HR or labor-relations review, since staff will be recorded too
- [ ] Door signage for visitors, and a camera-off switch patients and staff can use at any time
- [ ] Privacy office sign-off: encrypted hospital storage, access logging, a business associate agreement for any outside vendor or cloud
- [ ] Retention rule: raw video deleted after the audit window (e.g. 30 days); models trained only on CSI, skeletons and labels
- [ ] Hospital IT approval of the 2.4 GHz channel and the new transmitters
- [ ] Clinical engineering review of electrical safety: non-medical equipment within 1.5 m of the patient (the IEC 60601-1 patient environment) needs isolation or must move
- [ ] Infection control: wipeable enclosures that tolerate hospital disinfectants, and an infection control risk assessment permit for any above-ceiling work
- [ ] Cables in raceway, no trip hazards; skip behavioral-health units, where mounts and cables are ligature risks

If the system's outputs will drive clinical alerts, it is likely a medical device under FDA rules. Get regulatory advice before the pilot ends, not after the model works.

Data collection plan

Collect in four phases, moving from scripted actors in a simulation room to consented patients in real rooms. Real falls are rare (on the order of a few per 1,000 patient-days, approximate), so fall training data must come from simulation, and patient data is used mainly for occupancy, bed exits and false-alarm testing.

Phase	Where	Who	Goal	Target volume
0. Bench	Lab or office	Team	Firmware, packet rate, sync check under 50 ms	2 days of logs
1. Simulation	Hospital sim center or mock room, real hospital bed	10+ volunteer actors of varied size	Scripted protocol below, including falls onto mats	~30 h; 250+ falls, 500+ bed exits
2. Empty real rooms	Patient rooms between discharge and admission	Staff actors	Room diversity: shapes, beds, equipment	15+ rooms, 2 installs each
3. Patient rooms	Consented patients' rooms, 24/7	Patients, staff, visitors	Natural behavior, nights, visits	20–30 patients, 1,000+ patient-hours

Scripted protocol for phases 1 and 2 (one session is about 90 minutes):

Activity	Reps or duration	Variations
Empty room baseline	10 min	Door open/closed, curtain drawn/open, equipment moved
Lying still in bed	5 min each	On back, left side, right side; head of bed at 0°, 30°, 60°
Paced breathing (with chest belt)	2 min each	Slow, normal, fast

Activity	Reps or duration	Variations
Sit up to bed edge	10	Fast and slow, with and without bed rail
Bed exit, stand, walk to bathroom door, return	10	Each accessible side of the bed
Bed-to-chair transfer	5	With and without walker
Walking in room	3 min	Normal pace, shuffling gait
Staff visit	3 × 5 min	1–2 staff at bedside with actor in bed
Simulated falls onto crash mat	5 per type	Forward, backward, sideways, slide off bed, slow collapse
Fall-like non-falls	5 per type	Pick up object, sit on floor deliberately, fast sit into chair

Simulated falls need trained actors, a spotter and crash mats; the fall-like non-falls are the hard negatives that keep false alarms down.

Vary the environment on purpose. Within every phase, change bed height and position, run the air-mattress pump and IV pumps, open and close the bathroom door, and record at different times of day. Take the ESP32s down and remount them between sessions, since the deployed system will never be installed in exactly the same spot twice.

Preprocessing and features

Turn raw packets into fixed-size windows of cleaned amplitude and sanitized phase, normalized against an empty-room calibration recorded at every install. That calibration step is the cheapest defense against the room-to-room drift described in the overview.

1. **Parse.** Convert int8 I/Q pairs to complex values and drop guard-band, null and DC subcarriers (about 52 usable of 64 at HT20).
2. **Resample.** Put each receiver on a uniform 100 Hz grid using the ESP timestamps; interpolate gaps under 50 ms, flag longer ones, and drop sessions losing more than 5% of packets.

3. **Amplitude.** Take the magnitude, remove outliers with a Hampel filter, and normalize each packet across subcarriers, because the ESP32's automatic gain control rescales amplitude from packet to packet.
4. **Phase.** Raw phase is scrambled by clock and frequency offsets. Remove the linear slope and offset across subcarriers, then use the change in phase over time rather than its absolute value.
5. **Calibrate.** Record 5 minutes of empty room at each install and z-score all later data against it.
6. **Filter by task.** 0.1–0.7 Hz band for breathing; 0.3–20 Hz for body motion; a slow moving average for presence.
7. **Window.** 2 s windows (200 samples) with a 0.5 s step for activity and events; 30 s windows for breathing. Each window takes its majority label, and windows that straddle a transition are flagged.

Inputs for each model family. Classical models get hand-built features: per-subcarrier variance, the top principal components, a Doppler spectrogram from those components, cross-link correlation and RSSI statistics. Deep models get a tensor of 3 links × 52 subcarriers × 200 time steps, with amplitude and phase-change as two channels.

Model training plan

Start with simple baselines, then train one multi-task deep model, and judge everything on rooms and people it has never seen. A model that only works in its training room is the default outcome with CSI, so the evaluation splits matter more than the architecture.

1. **Baselines.** A moving-variance threshold for presence, and gradient-boosted trees (LightGBM) on the hand-built features for occupancy, in-bed and posture. These set the bar every deep model must beat.
2. **Multi-task deep model.** A CNN encoder over subcarriers × time, followed by a small temporal model (GRU or Transformer), with separate heads for occupancy, in-bed, posture and events. Falls use two stages: a sensitive detector on the live stream, then a classifier on each candidate window.
3. **Breathing.** Begin with signal processing: during stillness, find the spectral peak of the filtered principal component. Train a learned model only if that misses the target.
4. **Camera-to-CSI distillation.** Add an auxiliary head that predicts the camera's keypoints from CSI. Three single-antenna links will not recover a full skeleton, but this extra signal usually improves coarse posture.

5. **Pretraining on unlabeled data.** Phase 3 yields far more CSI than reviewed labels. Pretrain the encoder with masked reconstruction or contrastive learning on all of it, then fine-tune on labels.
6. **Generalizing to new rooms.** Use per-install calibration, augmentation (subcarrier dropout, link dropout, amplitude scaling, time warping), and test whether 10 minutes of labeled data from a new room is enough to fine-tune.

Evaluation splits. Never split windows at random: neighboring windows are near-duplicates, and scores come out inflated. Report every metric on four splits: leave-one-actor-out, leave-one-room-out, leave-one-install-out, and train-on-earlier-days, test-on-later-days.

Task	Primary metric	Starting target (agree with clinicians)
Room occupancy	Macro F1	⩾ 0.95
Patient in bed	F1; bed-exit detection delay	⩾ 0.95; alert within 5 s of standing
Pre-exit warning	Lead time from sitting on bed edge to standing	Median lead time > 0 s at ⩾ 90% recall
Posture	Macro F1 and confusion matrix	⩾ 0.85
Falls	Event-level recall at a fixed false-alarm rate	⩾ 90% recall at ⩽ 1 false alarm per room per day
Breathing rate	Mean absolute error vs chest belt	⩽ 2 breaths per minute

False alarms per room per day is the number clinicians will care about most, because alarm fatigue is already a known hospital safety problem. Track it for every task that raises an alert.

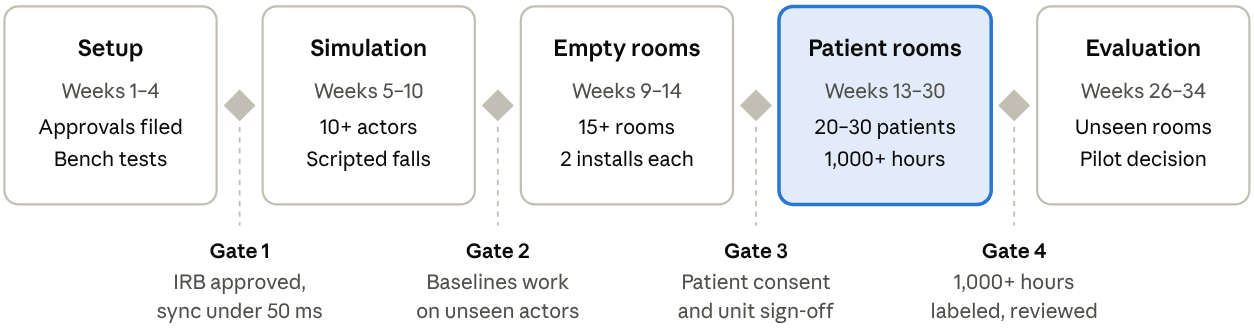
Deployment. Quantize the final model to int8. Run it either on the ESP32-S3 with Espressif's ESP-DL library or on a small hub per ward. Version datasets with DVC and log experiments in MLflow so each result traces back to its exact data.

Timeline, risks and costs

The plan runs about 34 weeks from filing approvals to a go/no-go decision on a camera-free pilot, with patient data starting around week 13. If the IRB takes longer, the simulation and empty-room phases simply run longer.

Patients join at week 13, after simulation and empty rooms prove the setup

Five phases, four gates · bands not to scale, weeks approximate



roadmap · 5 phases, 4 gates, ~34 weeks

Phases overlap by a few weeks so rooms and actors stay busy; each gate must pass before the next phase starts collecting in earnest.

Risk	Effect	Mitigation
Models fail in new rooms	Accuracy drops at deployment	Leave-one-room-out testing, per-install calibration, 15+ rooms of data
Slow IRB or privacy approval	Patient phase slips	File in week 1; run simulation phases meanwhile
Too few real falls	Fall model cannot be validated on patients	Train on simulated falls; report only false-alarm rate from patient data
2.4 GHz congestion from hospital Wi-Fi	Noisy CSI, dropped packets	IT-chosen channel, packet-loss alarms, ESP32-C5 at 5 GHz as fallback
Staff and visitor motion	False bed-exit and fall alerts	`others_present` label, staff-visit protocol, multi-person training data

Risk	Effect	Mitigation
Nodes bumped during cleaning	Silent drift in the data	Tamper-evident mounts, daily empty-room drift check
Camera occlusion by curtains or blankets	Missing or wrong labels	Depth camera, an explicit unknown label, human review

Item, per collection room	Qty	Rough unit cost (USD)
ESP32-S3-DevKitC-1U	4	~15
2.4 GHz antenna + u.FL pigtail	4	~5
Depth camera (Femto Bolt or RealSense D455 class)	1	~400–450
Mini PC or Raspberry Pi 5 with 1 TB SSD	1	~150–400
Powered USB hub, cables, mounts, wipeable enclosures	1 set	~100
Crash mats for the simulation phase (one-time)	2	~200

Prices are rough estimates from memory, not quotes. A collection room comes to roughly $800–1,100; a deployed camera-free room needs only the four ESP32s, antennas and enclosures, around $100, plus a shared ward hub.

Sources

- Espressif esp-csi repository: chip support, collection methods, antenna guidance, esp-crab and console_test
- Ultralytics YOLO26: pose model and license terms